

LIEGE AP, Belgium

WMO: 064780

Lat: 50.646N

Lon: 5.456E

Elev: 659

StdP: 14.35

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.9	23.4	13.4	11.2	23.7	17.9	13.9	25.9	30.1	47.4	27.6	46.7	6.4	50	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	85.8	68.3	81.7	66.6	78.1	65.3	70.0	81.9	68.2	78.6	66.4	75.9	7.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	98.1	75.8	64.3	92.6	73.0	62.8	87.7	71.0	34.4	82.1	32.8	78.8	31.4	76.0	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	21.3	18.9	DB	15.3	92.9	5.7	3.3	11.2	95.3	7.9	97.2	4.7	99.1	0.5	101.4
			WB	14.6	73.9	5.6	2.6	10.5	75.8	7.3	77.3	4.1	78.7	0.1	80.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	37.4	38.7	43.8	49.9	56.5	61.9	65.8	65.2	59.2	52.5	44.4	38.8	(6)
(7)		DBStd	12.12	7.52	7.56	6.84	7.25	6.85	6.32	6.41	5.79	5.71	6.56	6.57	7.30	(7)
(8)		HDD50	1618	393	318	212	89	17	0	0	0	1	49	188	351	(8)
(9)		HDD65	5260	856	735	657	454	276	136	69	68	192	388	617	812	(9)
(10)		CDD50	2074	1	3	20	87	219	356	488	470	278	127	21	4	(10)
(11)		CDD65	241	0	0	0	2	13	41	92	73	19	1	0	0	(11)
(12)		CDH74	1841	0	0	0	17	96	329	719	576	102	2	0	0	(12)
(13)		CDH80	522	0	0	0	1	18	85	220	182	16	0	0	0	(13)
(14)	Wind	WSAvg	9.0	10.8	10.4	9.9	8.5	8.1	7.8	7.6	8.1	9.2	9.6	10.6	(14)	
(15)	Precipitation	PrecAvg	32.6	2.8	2.5	2.4	2.0	2.5	3.0	3.2	2.5	2.6	2.6	3.0	(15)	
(16)		PrecMax	42.0	5.6	5.6	5.4	5.0	6.2	5.9	10.5	10.1	7.3	6.9	4.3	6.8	(16)
(17)		PrecMin	24.2	0.2	0.3	0.2	0.0	0.8	0.5	1.4	0.7	0.6	1.3	0.2	0.8	(17)
(18)		PrecStd	5.2	1.3	1.2	1.1	1.1	1.3	1.4	1.9	2.2	1.6	1.1	1.0	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	58.6	66.2	76.6	82.5	88.3	91.7	90.7	82.3	73.0	63.0	55.8	(19)
(20)			MCWB	50.9	48.2	53.3	59.8	64.0	69.8	69.0	70.2	66.7	62.0	55.6	51.8	(20)
(21)		2%	DB	51.9	53.8	61.3	70.8	76.1	81.9	86.0	84.9	76.9	67.9	58.5	53.5	(21)
(22)			MCWB	49.0	47.5	51.4	57.0	61.9	66.8	68.5	68.2	64.7	59.8	53.8	50.4	(22)
(23)		5%	DB	49.4	50.9	57.2	66.0	72.3	77.4	81.4	80.5	72.3	64.3	56.0	51.3	(23)
(24)			MCWB	46.9	47.0	49.9	54.8	60.2	65.2	66.5	66.6	62.5	58.2	52.3	48.8	(24)
(25)		10%	DB	47.1	48.7	53.7	61.5	68.3	73.6	77.5	76.0	68.4	61.5	53.5	48.8	(25)
(26)			MCWB	44.6	45.5	48.3	52.6	58.5	63.5	65.3	64.5	60.7	56.8	50.5	46.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	51.4	55.6	62.4	67.5	71.7	72.4	73.2	68.6	63.0	57.3	53.1	(27)
(28)			MCDWB	53.7	55.3	63.6	74.5	78.4	83.0	86.0	86.9	78.4	69.6	61.0	55.0	(28)
(29)		2%	WB	49.3	49.2	52.7	58.3	64.3	68.7	69.9	69.8	65.7	60.8	54.5	50.9	(29)
(30)			MCDWB	51.3	52.1	58.4	67.5	73.4	79.0	82.4	81.4	74.6	66.2	57.8	53.0	(30)
(31)		5%	WB	47.0	47.4	50.7	55.6	61.6	66.3	68.1	67.5	63.6	59.1	52.6	48.9	(31)
(32)			MCDWB	49.2	50.3	55.7	64.1	69.5	75.3	78.5	77.5	70.7	63.2	55.7	51.1	(32)
(33)		10%	WB	44.8	45.6	48.9	53.4	59.3	64.2	66.3	65.5	61.6	57.4	50.7	46.6	(33)
(34)			MCDWB	46.9	48.5	53.2	60.9	66.8	72.2	76.1	73.8	67.5	61.1	53.3	48.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	9.1	12.0	14.7	15.1	15.7	15.5	15.1	13.5	11.1	8.3	7.3	(35)
(36)			MCDBR	8.8	11.6	17.1	20.0	20.3	21.6	21.2	22.1	18.1	13.9	10.3	8.2	(36)
(37)			MCWBR	7.9	8.3	10.3	10.9	10.6	11.0	9.1	9.8	9.0	8.3	7.6	6.9	(37)
(38)		5% WB	MCDWR	8.8	10.1	14.3	18.0	17.9	19.0	19.2	19.3	16.4	12.5	9.4	8.1	(38)
(39)			MCWBR	8.1	8.0	9.5	10.6	10.2	10.9	9.2	9.8	9.1	8.1	7.5	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.313	0.336	0.368	0.401	0.410	0.410	0.412	0.405	0.376	0.353	0.328	0.311	(40)	
(41)		taud	2.427	2.383	2.306	2.238	2.252	2.269	2.261	2.307	2.373	2.433	2.430	2.398	(41)	
(42)		Ebn at Noon	228	252	262	266	269	269	266	262	258	242	219	208	(42)	
(43)		Edh at Noon	19	26	33	40	41	41	41	37	31	24	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	247	460	826	1311	1558	1697	1611	1372	1035	612	292	194	(44)	
(45)		RadStd	36	91	117	151	154	153	193	144	108	51	35	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page